Notes: Hombert and Matray (JF, 2018)

Can Innovation Help U.S. Manufacturing Firms Escape Import Competition from China?

Contents

1 Big picture

- **Question.** Does ex ante investment in R&D make U.S. manufacturing firms more resilient to import competition from China?
- **Setting.** The paper studies U.S. manufacturing firms during the rise of Chinese imports, combining Compustat firm data, UN Comtrade trade data, patent-based inventor-location data, state R&D tax-credit variation, and text-based product-differentiation measures.
- **Core challenge.** A simple correlation between R&D and resilience to trade shocks is not convincing because both R&D and import penetration are endogenous. More productive or better managed firms may both invest more in R&D and perform better; at the same time, weak domestic industries may attract more imports from China.
- **Main conceptual contribution.** Instead of asking whether firms *change* R&D after a trade shock, the paper directly estimates whether firms with more exogenous R&D *before* the shock are less harmed by import competition.
- **Main theoretical contribution.** The model separates two possibilities:
  - the *Arrow escape-competition effect*: innovation helps firms withstand competition, so the marginal return to innovation rises with trade pressure;
  - the *Schumpeterian erosion effect*: competition erodes the gains from innovation, so the marginal return to innovation falls with trade pressure.

  The sign of the interaction between import competition and innovation is therefore an empirical question.
- **Main empirical design.** The authors instrument China import penetration in the United States using Chinese import penetration in eight other high-income countries, and they instrument firm-level R&D using tax-induced changes in the user cost of R&D generated by staggered state R&D tax credits.
- **Main baseline result.** Rising Chinese import competition reduces sales growth and profitability, but the effect is much smaller for firms with larger stocks of R&D capital. Moving from the 25th to the 75th percentile of R&D stock roughly offsets half of the average sales-growth effect and about all of the average ROA effect.
- **Main mechanism result.** The protective role of R&D operates primarily through product differentiation rather than pure cost reduction. Firms with more R&D have more differentiated products, differentiation becomes more valuable when import competition increases, and high-R&D firms increase differentiation more when they are hit by trade shocks.
- **Main real-effects result.** Import competition leads firms to cut capital expenditures and employment, but these reductions are much smaller for R&D-intensive firms.
- **Economic takeaway.** Innovation helps U.S. manufacturing firms escape low-cost import competition mainly by allowing them to move away from head-to-head price competition and toward more differentiated products.

2 Data and measurement (key objects)

2.1 Why the China shock is a useful laboratory

- The paper exploits the surge in Chinese exports that followed China's transition toward a market-oriented economy and deeper integration into world trade.
- From 1991 to 2007, China's share of manufacturing imports rose from 6.7% to 25.0% in the United States and from 3.7% to 16.1% in a comparison group of other high-income countries.
- The cross-industry pattern is highly uneven: textiles, electronics, furniture, and industrial equipment see especially large increases in Chinese import penetration, whereas food, tobacco, petroleum, and printing see much smaller changes.

Interpretation. The setting gives the paper exactly what it needs: a large competition shock with strong cross-industry variation and a plausible supply-side origin in China.

2.2 Trade data and import-penetration measure

Trade data come from UN Comtrade and are aggregated to the four-digit SIC industry level for the United States and for eight other high-income countries: Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland.

China import penetration in U.S. industry j and year t is measured as

$$ImportPenetration_{jt}^{US} = \frac{ImportsFromChina_{jt}^{US}}{Employment_{j,1990}}. \tag{1}$$

The comparison-country measure is defined analogously:

$$ImportPenetration_{jt}^{HI} = \frac{ImportsFromChina_{jt}^{HI}}{Employment_{j,1990}}. \tag{2}$$

The denominator uses industry employment in 1990 rather than contemporaneous employment.

Interpretation. Using fixed pre-period employment avoids mechanically shrinking the denominator when import competition itself reduces domestic employment.

2.3 Instrument for import penetration

The import-penetration instrument comes from the classic China-shock design:

$$ImportPenetration_{jt}^{US} = a + b\, ImportPenetration_{jt}^{HI} + \eta_j + \tau_t + \varepsilon_{jt}, \tag{3}$$

where η_j are industry fixed effects and τ_t are year fixed effects.

The fitted value from this first-stage regression is the instrumented U.S. import-penetration variable used later in the second stage.

Table II reports a first-stage coefficient of about 1.36 on import penetration in other high-income countries, with a clustered standard error of about 0.12 and an F-statistic of 127.

Interpretation. The identifying idea is that the common component of Chinese export growth across destination countries reflects Chinese supply shocks rather than U.S.-specific demand or productivity shocks.

2.4 Firm-level R&D instrument

After the federal R&D tax credit was introduced in 1981, U.S. states gradually introduced their own state-level R&D tax credits. This creates variation in the user cost of R&D capital across states and over time.

Let z_{st} denote the tax-induced user cost of R&D in state s and year t. A firm's exposure depends on where it actually conducts R&D. Using inventor locations from patent records, the firm-specific user cost is

$$z_{it} = \sum_{s} w_{ist} z_{st}, \tag{4}$$

where w_{ist} is the 10-year moving-average share of firm i's inventors located in state s.

The first-stage flow regression is

$$\frac{R\&D_{it}}{Assets_{it}} = a + b\, z_{it} + \nu_i + \tau_t + \varepsilon_{it}, \tag{5}$$

with firm and year fixed effects.

Table III reports a coefficient of about -0.11 on the user cost of R&D, implying that a one-percentage-point decline in the user cost raises R&D expenditures by about 0.11% of assets. The implied price elasticity is about 1.3.

Interpretation. Firms whose inventors happen to be located in states that adopt more generous R&D tax credits face a lower effective price of R&D and therefore invest more in it for reasons plausibly unrelated to contemporaneous product-market shocks.

2.5 Constructing the stock of R&D capital

The paper needs a stock, not just a flow, of R&D. It therefore uses a perpetual-inventory rule:

$$R\&DStock_{it} = (1-\delta) R\&DStock_{i,t-1} + \widehat{R\&DFlow}_{it}, \tag{6}$$

where the depreciation rate is set to $\delta = 15\%$ and the flow is the *predicted* value from the first-stage user-cost regression.

Interpretation. The stock captures accumulated innovative capability rather than just contemporaneous spending. This is important because resilience to trade shocks is likely to depend on past investment in technology and product development.

2.6 Firm sample and main outcome variables

- The firm data come from Compustat and focus on U.S. manufacturing firms (SIC 2000–3999).
- Firms are required to have nonmissing assets and sales and at least three consecutive years of data.

- The core second-stage sample period is 1991–2007, matching the rise in China's import penetration.

The main outcomes are:

- **Sales growth,**
- **ROA,**
- **Capital expenditures divided by lagged property, plant, and equipment,**
- **Employment growth.**

All variables that can be positive or negative are winsorized at the 1% level in each tail; variables that are bounded below by zero are winsorized at the upper 1% tail.

2.7 Product differentiation

To open the mechanism black box, the paper uses the Hoberg–Phillips text-based product-space measure. For firm i in year t, product differentiation is

$$Differentiation_{it} = 1 - Similarity_{it}, \tag{7}$$

where similarity is based on overlap in the product descriptions of U.S. public firms' 10-K filings.

The firm-level measure used in the regressions is the average differentiation of a firm from its U.S. peers.

Interpretation. A higher value means the firm's product descriptions are more distinct from those of its competitors. This is the empirical proxy for the Sutton-style "quality ladder / differentiation" channel.

2.8 Descriptive facts

Table IV reports the main second-stage summary statistics:

- mean sales growth: 0.12,
- mean ROA: 0.023,
- mean capex/PPE: 0.38,
- mean employment growth: 0.063,
- mean import penetration: 0.0080 million dollars per worker,
- mean R&D stock / assets: 0.38,
- mean product differentiation: 0.973.

The cross-sectional standard deviation of import penetration is large relative to the mean, which is exactly what makes the interaction design economically informative.

Interpretation. The summary statistics already suggest two central facts: import exposure varies enormously across industries, and the stock of R&D is quantitatively important for many firms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Theoretical setup: innovation, competition, and the parameter of interest

Each firm chooses innovation effort $R \geq 0$ at cost

$$c(R) = (1+\theta)R + \frac{\rho}{2}R^2, \tag{8}$$

where θ captures the opportunity cost of R&D and $\rho > 0$ governs decreasing returns.

Let $I \in \{0, 1\}$ indicate whether innovation succeeds, with

$$\Pr(I = 1 \mid R) = R. \tag{9}$$

Firm performance when the innovation outcome is I and trade pressure is T is

$$\pi_I = a_I - b_I T, \qquad b_0 > 0,\ b_1 > 0. \tag{10}$$

This can be rewritten as

$$\pi = \alpha + \beta T + \gamma I + \delta T I, \tag{11}$$

where $\alpha = a_0$, $\beta = -b_0$, $\gamma = a_1 - a_0$, and $\delta = b_0 - b_1$.

Interpretation. The key parameter is δ.

- If $\delta > 0$, innovative firms are *less* sensitive to competition than noninnovators: the Arrow escape-competition effect dominates.
- If $\delta < 0$, competition erodes the value of innovation: the Schumpeterian effect dominates.

3.2 Why regressing R&D on competition does not identify the answer

From the model, firms choose R&D according to

$$R = \frac{1}{\rho}\big(\gamma + \delta T - (1+\theta)\big). \tag{12}$$

Interpretation. This equation explains why the standard "regress innovation on competition" approach is not enough.

- The slope identifies δ/ρ, not δ itself, so the magnitude of the Arrow-vs.-Schumpeter tradeoff is confounded with the cost curvature of R&D.
- Worse, if competition shocks tighten financing constraints or create agency frictions, then θ is endogenous to T, so even the sign of the coefficient can be misleading.

This is the paper's key methodological point: ex post R&D responses do *not* cleanly identify whether R&D protects firms from trade shocks.

3.3 Direct empirical approach

The paper instead estimates the direct interaction equation

$$\pi = \alpha + \beta T + \gamma R + \delta T R + \varepsilon, \tag{13}$$

instrumenting both T and R.

The panel implementation is

$$\begin{aligned} Y_{ijt} = \alpha &+ \beta\, ImportPenetration_{j,t-1} + \gamma\, R\&DStock_{i,j,t-1} \\ &+ \delta\big(ImportPenetration_{j,t-1} \times R\&DStock_{i,j,t-1}\big) + Controls_{i,j,t-1} + \nu_i + \omega_{jt} + \varepsilon_{ijt}. \end{aligned} \tag{14}$$

Here:

- Y_{ijt} is an outcome for firm i in industry j and year t,
- ν_i are firm fixed effects,
- ω_{jt} are industry-by-year fixed effects,
- controls include log assets, log age, and log age interacted with import penetration.

Interpretation. The coefficient of interest is again δ. A positive δ means that firms with higher R&D stock are less adversely affected by Chinese import competition.

3.4 Panel structure, lags, and standard errors

The paper uses lagged import penetration and lagged R&D stock. This is important both conceptually and econometrically.

- Conceptually, the paper wants to know whether *preexisting* innovative capability helps when the trade shock hits.
- Econometrically, lagging the right-hand-side variables weakens simultaneity concerns.

Because the key regressors are predicted from first stages, the standard errors are bootstrapped, with clustering at the industry-year level, to account jointly for:

- the uncertainty from the trade first stage,
- the uncertainty from the R&D first stage,
- the dependence structure in the second-stage panel.

Interpretation. The paper's precision does not come from one heroic exclusion restriction alone. It comes from a large panel, two sources of quasi-experimental variation, and inference that explicitly respects the generated-regressor structure.

3.5 Mechanism specifications: product differentiation

The paper uses three complementary mechanism regressions.

Channel 1: does R&D create more differentiated products?

$$Differentiation_{it} = a + b\, R\&DStock_{it} + Controls_{it} + \nu_i + \omega_t + \varepsilon_{it}. \tag{15}$$

Channel 1 continued: does differentiation matter more when import competition rises?

$$\begin{aligned} Y_{it} = a &+ \beta\, ImportPenetration_{jt} + \lambda\, Differentiation_{i,t-1} \\ &+ \phi\,(ImportPenetration_{jt} \times Differentiation_{i,t-1}) + Controls_{it} + \nu_i + \omega_t + \varepsilon_{it}. \end{aligned} \tag{16}$$

Channel 2: do high-R&D firms increase differentiation more when trade pressure rises?

$$\begin{aligned} Differentiation_{it} = a &+ \beta\, ImportPenetration_{jt} + \gamma\, R\&DStock_{it} \\ &+ \delta\,(ImportPenetration_{jt} \times R\&DStock_{it}) + Controls_{it} + \nu_i + \omega_{jt} + \varepsilon_{it}. \end{aligned} \tag{17}$$

Interpretation.

- A positive coefficient on $R\&DStock$ in the first regression says R&D is associated with more differentiated products.
- A positive coefficient on $ImportPenetration \times Differentiation$ says differentiation becomes more valuable when Chinese competition intensifies.
- A positive coefficient on $ImportPenetration \times R\&DStock$ in the differentiation regression says high-R&D firms respond to trade shocks by differentiating more.

3.6 Industry-heterogeneity mechanism test

The paper also tests whether R&D matters more in industries where differentiation is generally more important:

$$\begin{aligned} Y_{it} = a &+ \beta_1\big(ImportPenetration_{jt} \times R\&DStock_{it}\big) \\ &+ \beta_2\big(ImportPenetration_{jt} \times R\&DStock_{it} \times IndustryDiff_j\big) \\ &+ \text{all lower-order terms} + \varepsilon_{it}, \end{aligned} \tag{18}$$

where $IndustryDiff_j$ is a dummy equal to one when the industry's average differentiation is above the sample median.

Interpretation. If the whole story is really about differentiation, then the protective effect of R&D should be especially strong in industries where differentiation matters more in the first place.

3.7 Real-effects specifications

For capital expenditures and employment, the paper uses the same interaction design as in the baseline:

$$RealOutcome_{it} = a + \beta\, ImportPenetration_{jt} + \gamma\, R\&DStock_{it} + \delta\,(ImportPenetration_{jt} \times R\&DStock_{it}) + Controls \tag{19}$$

Interpretation. If R&D protects firm performance, it should also protect real investment and employment. The model predicts that high-R&D firms should downsize less because their effective productivity or product-market position erodes less.

3.8 Alternative trade-shock specification: PNTR

A major robustness test uses the U.S. grant of Permanent Normal Trade Relations (PNTR) to China in 2000. The key industry variable is

$$NTRGap_j = Non\text{-}NTRRate_j - NTRRate_j. \tag{20}$$

The alternative specification interacts this gap with a post-2000 dummy and with R&D stock:

$$Y_{ijt} = a + \beta\big(NTRGap_j \times Post_t\big) + \delta\big(NTRGap_j \times Post_t \times R\&DStock_{it}\big) + Controls_{it} + \nu_i + \omega_{jt} + \varepsilon_{ijt}. \tag{21}$$

Interpretation. Industries that faced a larger fall in expected trade barriers after PNTR experienced larger import shocks. If the main results are real, they should reappear under this completely different source of trade-shock variation.

4 Identification and instruments (what makes the estimates credible)

4.1 Why both import penetration and R&D must be instrumented

The paper is explicit that there are *two* separate endogeneity problems.

- **Import penetration is endogenous.** Weak domestic productivity can itself attract more imports.
- **R&D is endogenous.** Better firms may spend more on R&D because they have stronger opportunities, better managers, or stronger demand.

Interpretation. Instrumenting only the trade shock is not enough. If R&D remains endogenous, the interaction term is still contaminated by unobserved heterogeneity in innovative capability.

4.2 Why the China-shock instrument is credible

The comparison-country instrument works if the common rise in Chinese imports across destinations is driven by Chinese supply shocks such as:

- productivity growth in China,
- lower trade costs,
- WTO-related integration,
- and expansion of Chinese manufacturing capacity.

The key empirical fact is that the same industries that see import surges from China in the United States also see them in other high-income countries.

Interpretation. The instrument does not require that China create all value added itself. It only requires that the increase in Chinese exports be driven by shocks originating on the Chinese side rather than by destination-country demand alone.

4.3 Why the R&D-tax-credit instrument is credible

State R&D tax credits create changes in the effective price of R&D that vary over time and across firms depending on where the firm's inventors are located. This gives the paper a source of within-firm variation in innovative investment that is unrelated to the firm's current import exposure.

The paper addresses the obvious concern—that states may adopt R&D credits in response to local economic weakness or strength—by referring to prior literature and by showing in the appendix that changes in economic conditions do not predict R&D policy changes.

Interpretation. The exclusion restriction is never literally testable, but the design is much more convincing than simply using raw R&D expenditures and hoping that firm fixed effects solve everything.

4.4 Why firm fixed effects and industry-year fixed effects matter

The preferred specification includes:

- **firm fixed effects** to absorb time-invariant firm quality, management, product mix, and all other fixed firm characteristics;
- **industry-year fixed effects** to absorb industry-specific productivity shocks, changes in demand, and any common shock to all firms in an industry-year.

Interpretation. With these fixed effects, the interaction coefficient is identified by comparing firms with different *predicted* R&D stocks *within the same industry and year*. That is a much tougher and cleaner comparison than a raw cross-section.

4.5 Why the geographic-dispersion assumption is plausible

Because the R&D instrument comes from variation in the states where firms conduct R&D, the design needs firms within a given industry-year to be spread across states. The paper checks this using inventor locations.

Table V shows that, on average across industries, the top state accounts for about 32.9% of an industry's R&D, the second state for 17.2%, the third for 11.3%, the fourth for 8.2%, and the fifth for 6.1%.

Interpretation. Industries are somewhat clustered, but not so concentrated that the state-level R&D-tax-credit variation disappears within industry-years.

4.6 Alternative explanations the paper tries to rule out

The paper runs a particularly broad set of alternative-explanation tests.

- **Correlated demand shocks across rich countries.** This is the main residual threat to the import-penetration instrument. The paper cannot eliminate it completely, but argues the industry patterns are far more consistent with Chinese supply shocks.

- **Firm age effects.** Because predicted R&D stock may mechanically covary with age, the paper includes age interacted with import penetration.
- **Endogenous relocation to R&D-credit states.** The paper uses a 10-year moving average of inventor shares and also checks robustness to initial-location measures.
- **California dominance.** The appendix excludes firms with more than half of their R&D activity in California and finds similar results.
- **Intermediate-input effects.** The paper adjusts import exposure for input-market shocks and also estimates separate effects for final-good and input-market competition.

Interpretation. A striking feature of the paper is that the robustness exercises are not cosmetic. Each one is tied to a specific alternative mechanism.

4.7 What the paper can and cannot distinguish

The paper is very persuasive on *whether* R&D protects firms from Chinese competition. It is less definitive on the exact deeper reason.

- It strongly supports a product-differentiation mechanism.
- It is less able to say exactly how much of that comes from higher quality, broader product scope, faster adaptation, or other innovation channels.
- It does not speak to general-equilibrium effects. If every U.S. firm had more R&D, would all of them keep market share, or would they just take it from one another?

Interpretation. The paper is strongest on the micro-level interaction between exogenous R&D and exogenous trade pressure, not on the macro counterfactual.

4.8 Relation to the literature

The paper connects three literatures.

- **Import competition and firms.** It extends work showing that imports hurt domestic output, profits, investment, and employment by asking *which* firms cope better.
- **Innovation and competition.** It complements the literature that studies whether competition raises or lowers ex post innovation by instead directly estimating whether preexisting R&D shields firms from competition.
- **R&D policy.** It gives a concrete channel through which R&D subsidies may matter: not just more innovation, but greater resilience to trade shocks.

5 Tables and figures

The paper's empirical evidence is spread across figures and tables that line up very closely with the logic of the paper: first define the trade shock, then validate the two instruments, then show the main performance results, then open the differentiation mechanism, and finally trace the real effects and robustness.

5.1 Figure 1 and Table I: the China shock is large and highly uneven across industries

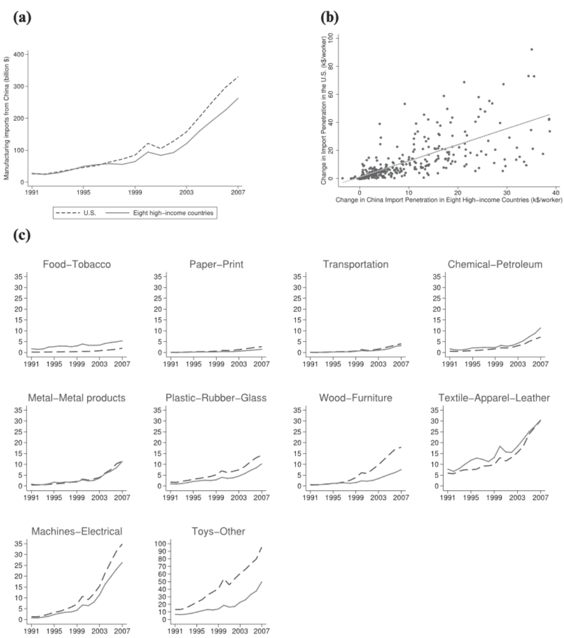

Figure 1. Manufacturing imports from China to the United States and to other high-income countries. Panel A plots total manufacturing imports (in 2007 billion USD) from China to the United States (blue line) and to a group of eight other high-income countries (red line: Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland). Panel B plots China's import penetration in the United States (blue line) and in a group of eight other high-income countries (red line; Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland) by broad manufacturing industry. For each broad industry, import penetration is measured as imports (in 2007 k$) from China in the industry divided by industry employment in 1990. Panel C plots the 1991 to 2007 change in China's import penetration in the United States on the y-axis against China's import penetration in a group of eight other high-income countries (Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland) on the x-axis. Each dot represents a four-digit manufacturing industry. For each industry, change in import penetration is measured as the change in imports (in 2007 k$) from China in the industry from 1991 to 2007 divided by industry employment in 1990.

Table I

Change in China's Import Penetration by Broad Industry

The table ranks two-digit manufacturing industries in descending order of the change in China's import penetration in the United States. For each industry, the change in import penetration is measured as the change in imports (in 2007 k$) from China in the industry from 1991 to 2007 divided by industry employment in 1990.

Two-Digit SIC Industries		1991 to 2007 Change in China's Import Penetration (k$/worker)
31	Leather and leather products	103.5
39	Miscellaneous manufacturing industries	82.2
36	Electronic and other electronic equipment	50.4
25	Furniture and fixtures	34.0
35	Industrial machinery and equipment	33.0
23	Apparel and other textile products	28.0
33	Primary metal industries	13.8
30	Rubber and miscellaneous plastics products	13.4
32	Stone, clay, and glass products	10.9
34	Fabricated metal products	9.6
38	Instruments and related products	7.8
28	Chemicals and allied products	7.1
24	Lumber and wood products	5.6
26	Paper and allied products	5.3
37	Transportation equipment	3.9
22	Textile mill products	2.9
29	Petroleum and coal products	2.6
20	Food and kindred products	1.8
27	Printing and publishing	1.5
21	Tobacco products	0.2

Read:

- Figure 1 shows that manufacturing imports from China rise sharply over time in both the United States and the comparison group of high-income countries.
- The industry panels in Figure 1 show that the rise is very concentrated: textiles, electronics, furniture, and machinery experience especially large increases in Chinese penetration.
- Table I ranks broad industries by the 1991–2007 increase in Chinese import penetration. Leather products, miscellaneous manufacturing, electronics, furniture, machinery, and apparel are at the top; tobacco, printing, food, and petroleum are at the bottom.

Economic message. The trade shock is both *big* and *selective*. The paper's design needs exactly this kind of cross-industry variation because resilience can only be studied when some industries are hit much harder than others.

5.2 Table II and Figure 2: the two instruments are strong

Table II

Instrument for Import Penetration: First-Stage Regression

The sample comprises four-digit industries over the 1991 to 2007 period. We estimate a linear regression model where the dependent variable is China's import penetration in the United States at the industry-year level. The dependent variable is China's import penetration in a group of eight other high-income countries (Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland). Import penetration is measured as imports (in 2007 k$) from China divided by industry employment in 1990. The regression includes industry and year fixed effects. Standard errors are clustered by industry and year. *** indicates statistically different from zero at the 1% level of significance.

	Import Penetration in the United States
Import penetration in other high-income countries	1.36***
	(0.12)
Industry FE	Yes
Year FE	Yes
Observations	2,885
Adjusted-R^2	0.94

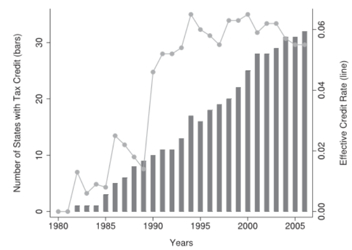

Figure 2. Number and average value of state R&D tax credits in the United States. Blue bars represent the number of U.S. states with R&D tax credits (left scale). Red dots plot the average effective R&D tax credit rate across the 50 states (right scale).

Read:

- Table II reports the first stage for import penetration. The coefficient on Chinese imports to the eight other high-income countries is about 1.36, with a standard error of about 0.12 and an F-statistic of 127.
- Figure 2 shows the diffusion of state R&D tax credits over time. Both the number of states offering credits and the average effective credit rate rise strongly over the sample period.

Economic message. Table II and Figure 2 do the heavy lifting for identification. The first validates the external trade-shock instrument; the second shows there is meaningful time-series and cross-state variation in the cost of R&D.

5.3 Tables III–V: the R&D first stage and the identifying sample

Table III

Instrument for R&D: First-Stage Regression

The sample comprises U.S. manufacturing firms over the 1973 to 2006 period from Compustat. We estimate a linear regression model on a firm-year panel where the dependent variable is R&D expenses over total assets. User cost of R&D at the firm-year level is calculated as the weighted average of state-year-specific user of cost of R&D to which the firm is eligible, where weights are based on the 10-year moving average of the share of the firm's inventors located in each state and the user cost of R&D at the state-year level comes from Wilson (2009) and is calculated using the Hall-Jorgenson formula as explained in Footnote 11. The regression includes firm and year fixed effects. Standard errors are clustered by industry. *** indicates statistically different from zero at the 1% level of significance.

	R&D
User cost of R&D	−0.11***
	(0.03)
Firm FE	Yes
Year FE	Yes
Observations	55,541
Adjusted-R^2	0.75

Table IV

Summary Statistics

The table reports summary statistics for the sample of U.S. manufacturing firms over the 1991 to 2007 period that we use in our second-stage regressions.

	Mean	Std.Dev.	25th	50th	75th	N
Sales growth	0.12	0.34	−0.032	0.079	0.23	24,753
ROA	0.023	0.37	−0.0095	0.12	0.2	25,424
Capex/PPE	0.38	0.53	0.12	0.22	0.41	25,210
Employment growth	0.063	0.26	−0.055	0.026	0.14	24,015
Import penetration (M$/worker)	0.0080	0.022	0.0000	0.0012	0.0056	24,598
R&D Stock/Total assets	0.38	0.62	0.042	0.17	0.43	25,494
Product differentiation	0.973	0.016	0.968	0.977	0.984	16,387

Table V

Geographic Distribution of R&D by Industry

For each industry, we rank states based on the share of R&D conducted in the state. The table reports the average share for each rank across all industries. Reading: The average industry has 32.9% of its R&D activity in the top state, 17.2% in the second state, and so on.

Industry-by-Industry Ranking of States	Share of Industry R&D in the State
Top state	32.9%
Second state	17.2%
Third state	11.3%
Fourth state	8.2%
Fifth state	6.1%
Sixth state	4.7%
Seventh state	3.6%
Eighth state	2.9%
Ninth state	2.4%
10th state	2.0%

Read:

- Table III shows that the user cost of R&D has a negative and statistically significant first-stage effect on R&D expenditures. The coefficient is about −0.11.
- Table IV reports the main summary statistics for the second-stage sample. Mean R&D stock is large relative to assets, and there is substantial dispersion in both import exposure and product differentiation.
- Table V shows that industries are geographically dispersed enough in their R&D activity for within-industry-year variation in state tax credits to matter.

Economic message. These tables are about *feasibility*. They show that the R&D instrument moves actual R&D spending, that the key variables vary enough to estimate interactions, and that the identification is not killed by extreme geographic clustering.

5.4 Tables VI and VII: the baseline result on sales growth and profitability

Table VI

R&D Capital in Import-Competing Industries: Effect on Sales Growth

The sample comprises U.S. manufacturing firms over the 1991 to 2007 period from Compustat. We estimate a linear regression model on a firm-year panel where the dependent variable is sales growth. All specifications include firm fixed effects, year fixed effects in columns (1) and (2), industry-by-year fixed effects in columns (3) and (4), log of total assets, and log of firm age as controls. *ImportPenetration* is the industry-year-level import penetration from China in the United States instrumented using China's import penetration in eight other high-income markets. *R&DStock* is the firm-year-level predicted stock of R&D capital instrumented using firm-specific tax-induced user cost of R&D capital. Standard errors are bootstrapped within industry-year clusters and reported in parentheses. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

	Sales Growth			
	(1)	(2)	(3)	(4)
Import penetration	−0.84***	−1.30***		
	(0.21)	(0.24)		
Import penetration × R&D Stock		0.83**	1.07***	1.11***
		(0.33)	(0.40)	(0.39)
Assets	0.01	0.03***	0.03***	0.03***
	(0.01)	(0.01)	(0.01)	(0.01)
Age	−0.20***	−0.23***	−0.23***	−0.23***
	(0.01)	(0.01)	(0.02)	(0.02)
R&D Stock		0.07***	0.07***	0.07***
		(0.02)	(0.03)	(0.03)
Import penetration × Age				−0.67
				(0.46)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	–
Industry-Year FE	No	No	Yes	Yes
Observations	23,907	23,907	23,907	23,907
R^2	0.24	0.24	0.34	0.34

Table VII

R&D Capital in Import-Competing Industries: Effect on Profitability

The sample comprises U.S. manufacturing firms over the 1991 to 2007 period from Compustat. We estimate a linear regression model on a firm-year panel where the dependent variable is ROA. All specifications include firm fixed effects, year fixed effects in columns (1) and (2), industry-by-year fixed effects in columns (3) and (4) and log of total assets, and log of firm age as controls. *ImportPenetration* is the industry-year-level import penetration from China in the United States instrumented using China's import penetration in eight other high-income markets. *R&DStock* is the firm-year-level predicted stock of R&D capital instrumented using firm-specific tax-induced user cost of R&D capital. Standard errors are bootstrapped within industry-year clusters and reported in parentheses. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

	ROA			
	(1)	(2)	(3)	(4)
Import penetration	−0.49**	−1.06***		
	(0.20)	(0.22)		
Import penetration × R&D Stock		1.13**	1.41***	1.42***
		(0.47)	(0.54)	(0.54)
Assets	0.06***	0.01	0.01	0.01
	(0.01)	(0.01)	(0.01)	(0.01)
Age	0.06***	0.13***	0.15***	0.15***
	(0.01)	(0.02)	(0.02)	(0.02)
R&D Stock		−0.19***	−0.21***	−0.21***
		(0.02)	(0.03)	(0.03)
Import penetration × Age				−0.20
				(0.34)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	–
Industry-Year FE	No	No	Yes	Yes
Observations	24,533	24,533	24,533	24,533
R^2	0.68	0.68	0.72	0.72

Read:

- Table VI shows that a one-standard-deviation increase in import penetration reduces annual sales growth by about 1.8 percentage points on average.
- The interaction between import penetration and R&D stock is positive and significant. In the preferred specification with industry-year fixed effects, moving from the 25th to the 75th percentile of R&D stock offsets the decline in annual sales growth by about 0.9 percentage points, roughly half of the unconditional effect.
- Table VII shows a similar pattern for profitability. A one-standard-deviation increase in import penetration reduces ROA by about 1.1 percentage points on average.
- Again the interaction with R&D stock is positive and significant. Moving from the 25th to the 75th percentile of R&D stock offsets the decline in ROA by roughly 1 percentage point, which is about the same size as the unconditional effect.

Economic message. Tables VI and VII are the heart of the paper. Chinese competition hurts the average U.S. manufacturing firm, but it hurts high-R&D firms much less. This is exactly what the direct interaction design was built to test.

5.5 Tables VIII–X: why the mechanism looks like product differentiation

Read:

- Table VIII supports **Channel 1**. More R&D leads to more differentiated products, and the performance benefit of differentiation becomes stronger when import competition rises. The interaction between import penetration and lagged differentiation is positive for both sales growth and ROA.

Table VIII

Product Differentiation Becomes More Important (Channel 1)

The sample comprises U.S. manufacturing firms over the 1996 to 2011 period from Compustat. We estimate a linear regression model on a firm-year panel where the dependent variable is average product differentiation (one minus the Hoberg and Phillips (2016) product similarity index) from U.S. peers in columns (1) and (2), sales growth in columns (3) and (4), and ROA in columns (5) and (6). All specifications include firm fixed effects, year fixed effects, log of total assets, and log of firm age as controls. Specifications in columns (2), (4), and (6) also include industry-by-year fixed effects. *ImportPenetration* is the industry-year-level import penetration from China in the United States instrumented using China's import penetration in eight other high-income markets. *R&DStock* is the firm-year-level predicted stock of R&D capital instrumented using firm-specific tax-induced user cost of R&D capital. Standard errors are clustered at the industry-year level and reported in parentheses. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

	Differentiation		Sales Growth		ROA	
	(1)	(2)	(3)	(4)	(5)	(6)
R&D Stock	0.0015**	0.0011*				
	(0.00067)	(0.00062)				
Import penetration			−16**		−9.7***	
			(7.2)		(3.7)	
Import penetration × Differentiation ($t-1$)			16**	15*	9.4**	12***
			(7.4)	(9.2)	(3.9)	(4.5)
Assets	0.00043**	−0.00044**	0.016	0.015	0.065***	0.07***
	(0.00022)	(0.00021)	(0.012)	(0.012)	(0.0088)	(0.0096)
Age	0.00031	0.00072	−0.21***	−0.2***	0.027*	0.027
	(0.00047)	(0.00046)	(0.031)	(0.031)	(0.015)	(0.016)
Differentiation ($t-1$)			−0.51	−0.73	0.02	−0.069
			(0.37)	(0.46)	(0.17)	(0.21)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	–	Yes	–	Yes	–
Industry-Year FE	–	Yes	–	Yes	–	Yes
Observations	15,637	15,637	12,767	12,767	12,767	12,767
R^2	0.83	0.84	0.25	0.35	0.74	0.77

Table IX

Product Differentiation Increases (Channel 2)

The sample comprises U.S. manufacturing firms over the 1996 to 2011 period from Compustat. We estimate a linear regression model on a firm-year panel where the dependent variable is average product differentiation (one minus the Hoberg and Phillips (2016) product similarity index) from U.S. peers. All specifications include firm fixed effects, year fixed effects in columns (1) and (2), industry-by-year fixed effects in columns (3) and (4), log of total assets, and log of firm age as controls. *ImportPenetration* is the industry-year-level import penetration from China in the United States instrumented using China's import penetration in eight other high-income markets. *R&DStock* is firm-year-level predicted stock of R&D capital instrumented using firm-specific tax-induced user cost of R&D capital. Standard errors are clustered at the industry-year level and reported in parentheses. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

	Product Differentiation			
	(1)	(2)	(3)	(4)
Import penetration	0.035*	0.024		
	(0.02)	(0.018)		
Import penetration × R&D Stock		0.026	0.023**	0.023**
		(0.017)	(0.011)	(0.011)
Assets	−0.00082***	−0.00021	−0.00029	−0.00029
	(0.00026)	(0.00034)	(0.00034)	(0.00034)
Age	0.001**	0.00034	0.00058	0.0004
	(0.00052)	(0.00058)	(0.00059)	(0.00059)
R&D Stock		0.0018**	0.00087	0.00086
		(0.00088)	(0.00073)	(0.00073)
Import penetration × Age				−0.019
				(0.022)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	–
Industry-Year FE	No	No	Yes	Yes
Observations	15,896	15,896	15,896	15,896
R^2	0.78	0.78	0.83	0.83

Table X

R&D Matters More in Industries with High Differentiation

The sample comprises U.S. manufacturing firms over the 1996 to 2011 period from Compustat. We estimate a linear regression model on a firm-year panel where the dependent variable is sales growth in columns (1) and (2) and ROA in columns (3) and (4). All specifications include firm fixed effects, year fixed effects, log of total assets, and log of firm age as controls. Specifications in columns (2) and (4) also include industry-by-year fixed effects. *ImportPenetration* is the industry-year-level import penetration from China in the United States instrumented using China's import penetration in eight other high-income markets. *R&DStock* is the firm-year-level predicted stock of R&D capital instrumented using firm-specific tax-induced user cost of R&D capital. *IndustryDifferentiation* is a dummy variable equal to one if the average product differentiation (one minus the Hoberg and Phillips (2016) product similarity index) of the industry is above the median of the sample distribution. All specifications include all the double and triple interactions between these three variables; noninteracted terms are also included but not reported. Standard errors are clustered at the industry-year level and reported in parentheses. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

	Sales Growth		ROA	
	(1)	(2)	(3)	(4)
Import penetration × R&D Stock	0.25	0.48	0.01	0.21
	(0.35)	(0.40)	(0.62)	(0.64)
Import penetration × R&D Stock	1.55*	1.88**	2.43***	3.13***
× Industry differentiation	(0.90)	(0.97)	(0.92)	(1.00)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	–	Yes	–
Industry-Year FE	–	Yes	–	Yes
Observations	23,074	23,074	23,710	23,710
R^2	0.25	0.33	0.69	0.73

- Table IX supports **Channel 2**. Import penetration on its own tends to increase differentiation, and the interaction between import penetration and R&D stock is positive: high-R&D firms differentiate more when the China shock intensifies.
- Table X shows that the protective role of R&D is stronger in industries where product differentiation is more important. The triple interaction between import penetration, R&D stock, and high-industry-differentiation is positive and significant for both sales growth and ROA.

Economic message. The mechanism evidence is unusually coherent. R&D does not just make firms *generally better*; it makes them better in exactly the dimension that theory says should matter most against low-cost foreign competition—product differentiation.

5.6 Tables XI and XII: real effects on capital expenditures and employment

Table XI

R&D Capital in Import-Competing Industries: Effect on Capital Expenditures

The sample comprises U.S. manufacturing firms over the 1991 to 2007 period from Compustat. We estimate a linear regression model on a firm-year panel where the dependent variable is capital expenditures divided by lagged property, plant, and equipment. All specifications include firm fixed effects, year fixed effects in columns (1) and (2), industry-by-year fixed effects in columns (3) and (4), log of total assets, and log of firm age as controls. *ImportPenetration* is the industry-year-level import penetration from China in the United States instrumented using China's import penetration in eight other high-income markets. *R&DStock* is the firm-year-level predicted stock of R&D capital instrumented using firm-specific tax-induced user cost of R&D capital. Standard errors are clustered at the industry-year level and reported in parentheses. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

	Capital Expenditures			
	(1)	(2)	(3)	(4)
Import penetration	−0.74*	−1.65***		
	(0.39)	(0.45)		
Import penetration × R&D Stock		1.67***	1.77***	1.79***
		(0.60)	(0.68)	(0.67)
Assets	0.03***	0.04***	0.05***	0.05***
	(0.01)	(0.02)	(0.02)	(0.02)
Age	−0.41***	−0.42***	−0.44***	−0.44***
	(0.02)	(0.03)	(0.03)	(0.03)
R&D Stock		0.01	0.02	0.02
		(0.04)	(0.04)	(0.04)
Import penetration × Age				−0.33
				(0.82)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	–
Industry-Year FE	No	No	Yes	Yes
Observations	24,321	24,321	24,321	24,321
R^2	0.34	0.34	0.41	0.41

Table XII

R&D Capital in Import-Competing Industries: Effect on Employment

The sample comprises U.S. manufacturing firms over the 1991 to 2007 period from Compustat. We estimate a linear regression model on a firm-year panel where the dependent variable is employment growth. All specifications include firm fixed effects, year fixed effects in columns (1) and (2), industry-by-year fixed effects in columns (3) and (4), log of total assets, and log of firm age as controls. *ImportPenetration* is the industry-year-level import penetration from China in the United States instrumented using China's import penetration in eight other high-income markets. *R&DStock* is the firm-year-level predicted stock of R&D capital instrumented using firm-specific tax-induced user cost of R&D capital. Standard errors are clustered at the industry-year level and reported in parentheses. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

	Employment Growth			
	(1)	(2)	(3)	(4)
Import penetration	−0.18	−0.63**		
	(0.21)	(0.28)		
Import penetration × R&D Stock		0.77**	0.88**	0.92**
		(0.32)	(0.39)	(0.37)
Assets	0.02***	0.06***	0.07***	0.07***
	(0.01)	(0.01)	(0.01)	(0.01)
Age	−0.20***	−0.25***	−0.24***	−0.25***
	(0.01)	(0.01)	(0.02)	(0.02)
R&D Stock		0.13***	0.13***	0.13***
		(0.02)	(0.02)	(0.02)
Import penetration × Age				−0.81*
				(0.47)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	–
Industry-Year FE	No	No	Yes	Yes
Observations	23,197	23,197	23,197	23,197
R^2	0.24	0.25	0.36	0.36

Read:

- Table XI shows that a one-standard-deviation increase in import penetration reduces capital expenditures by about 1.6 percentage points of fixed assets on average.

- The interaction with R&D stock is again positive and strongly significant. Moving from the 25th to the 75th percentile of R&D stock offsets about 1.4 percentage points of the decline in capital expenditures.
- Table XII studies employment growth. Low-R&D firms cut employment much more than high-R&D firms when import competition rises.
- The coefficient estimates imply that firms at the bottom quartile of R&D reduce employment growth by about 1.3 percentage points after a one-standard-deviation trade shock, whereas firms at the top quartile see only a modest and statistically insignificant reduction.

Economic message. The performance results are not just accounting outcomes. They show up in real decisions about investment and jobs. High-R&D firms do not merely protect margins; they avoid some of the downsizing that import competition otherwise triggers.

5.7 Tables XIII and XIV: the robustness checks are economically important, not cosmetic

Table XIII

Alternative Instrument for Import Competition

The sample comprises U.S. manufacturing firms over the 1991 to 2007 period from Compustat. We estimate a linear regression model on a firm-year panel. All specifications include firm fixed effects, year fixed effects, log of total assets, and log of firm age as controls. Specifications in even-numbered columns also include industry-by-year fixed effects. *NTRGap* is the difference between NTR tariff and non-NTR tariff at the industry level. *Post* is a dummy equal to one from year 2000 onward. *R&DStock* is the firm-year-level predicted stock of R&D capital instrumented using firm-specific tax-induced user cost of R&D capital. Standard errors are clustered at the industry-year level and reported in parentheses. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

	Sales Growth		ROA		Capital Expenditures		Employment Growth	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
NTR Gap × Post	−0.12**		−0.10***		−0.14**		−0.05	
	(0.06)		(0.03)		(0.06)		(0.04)	
NTR Gap × Post × R&D Stock		0.17***		0.09**		0.31***		0.15***
		(0.05)		(0.04)		(0.06)		(0.04)
Assets	0.03***	0.03***	0.01	0.00	0.04***	0.04***	0.06***	0.07***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Age	−0.24***	−0.23***	0.06***	0.08***	−0.33***	−0.32***	−0.24***	−0.22***
	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.02)	(0.01)	(0.01)
R&D Stock	0.08***	0.05*	−0.14***	−0.18***	0.03	−0.03	0.15***	0.12***
	(0.02)	(0.03)	(0.02)	(0.02)	(0.03)	(0.03)	(0.02)	(0.02)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	–	Yes	–	Yes	–	Yes	–
Industry-Year FE	–	Yes	–	Yes	–	Yes	–	Yes
Observations	23,471	23,471	23,471	23,471	23,471	23,471	22,234	22,234
R^2	0.24	0.33	0.71	0.75	0.35	0.42	0.24	0.34

Table XIV

Controlling for Import Competition in Input Markets

The sample comprises U.S. manufacturing firms over the 1991 to 2007 period from Compustat. In columns (1) and (4), we estimate the same regression as in column (4) of Tables VI and VII except that we now use input-adjusted import competition defined as import competition in the final market minus import competition in intermediate input markets. In columns (2) and (5), we estimate the same regression as in column (1) of Tables VI and VII except that we now use both import competition in the final market and import competition in intermediate input markets. In columns (3) and (6), we estimate the same regression as in column (4) of Tables VI and VII except that we now use both import competition in the final markets and import competition in intermediate input markets. Standard errors are bootstrapped within industry-year clusters and reported in parentheses. *, **, and *** indicate statistically different from zero at the 10%, 5%, and 1% level of significance, respectively.

	Sales Growth			ROA		
	(1)	(2)	(3)	(4)	(5)	(6)
Net import penetration × R&D Stock	1.31***			1.74***		
	(0.47)			(0.62)		
Final market import penetration		−0.84***			−0.71***	
		(0.24)			(0.27)	
Input market import penetration		0.24			1.65*	
		(0.93)			(1.02)	
Final good import penetration × R&D Stock			1.20**			1.67**
			(0.53)			(0.69)
Input market import penetration × R&D Stock			−0.24			−1.23
			(2.14)			(2.69)
Assets	0.03***	0.01	0.03***	0.01	0.06***	0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Age	−0.23***	−0.20***	−0.23***	0.15***	0.06***	0.15***
	(0.02)	(0.01)	(0.02)	(0.02)	(0.01)	(0.02)
R&D Stock	0.08***		0.07***	−0.21***		−0.21***
	(0.03)		(0.03)	(0.03)		(0.03)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	–	Yes	–	–	Yes	–
Industry-Year FE	Yes	No	Yes	Yes	No	Yes
Observations	23,843	23,843	23,843	24,461	24,461	24,461
R^2	0.33	0.24	0.33	0.72	0.68	0.72

Read:

- Table XIII replaces the import-penetration instrument with PNTR-based variation. Industries with larger NTR gaps experience worse sales growth, profitability, capital expenditures, and employment after 2000, but these effects are again significantly attenuated for firms with more R&D stock.
- Table XIV controls for import competition in intermediate-input markets. The interaction between final-good import competition and R&D remains positive and significant for sales growth and ROA, while the interaction for input-market competition is small and insignificant.

Economic message. These tables matter because they attack two of the most serious alternative explanations:

- that the main results are an artifact of one specific trade-shock instrument,
- or that they simply capture firms' exposure to cheaper imported inputs rather than competition in output markets.

The core conclusion survives both tests.

6 Short summary

- The paper asks whether firms that invested more in R&D before the China shock were better able to withstand import competition.
- It instruments Chinese import penetration with Chinese imports to other rich countries and instruments firm-level R&D with state R&D-tax-credit variation.
- Higher Chinese import competition lowers sales growth, profitability, investment, and employment on average.
- These effects are significantly smaller for firms with larger exogenous stocks of R&D capital.
- The mechanism evidence points strongly to product differentiation: high-R&D firms are more differentiated, differentiation matters more when trade pressure rises, and high-R&D firms differentiate more after the shock.
- The results are robust to an alternative PNTR-based trade-shock instrument and to controls for input-market competition.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that R&D-intensive U.S. manufacturing firms were materially more resilient to import competition from China. This is not simply a raw correlation: the result survives instruments for both the trade shock and the stock of R&D, firm fixed effects, industry-year fixed effects, and a large set of robustness exercises. The interaction between import competition and R&D is positive for sales growth, profitability, capital expenditures, and employment.

7.2 Economic intuition

The paper's central economic idea is that R&D helps firms move away from pure price competition. If Chinese producers enter with very low costs, firms that only compete on cost are squeezed. But firms that used R&D to improve quality, product scope, or distinctiveness can preserve demand and margins more easily. This is why the data line up much better with a vertical-differentiation story than with a pure productivity story.

7.3 Takeaway

The deepest takeaway is not just that innovation is good. It is more specific: *the type of innovative capital that helps against low-wage import competition is the type that supports product differentiation.* That is why the paper is an important bridge between the trade literature, the innovation literature, and the corporate-finance literature on intangible investment.